

VISUAL BASIC 6.0 & ORACLE 10g INTEGRATION

Complete Laboratory Manual: Table Query, Form Setup, Property Settings, ADODB Connection, VB6 Coding & Output Windows

Development Environment	Visual Basic 6.0 (Enterprise / Professional Edition)
Database Management System	Oracle Database 10g Express Edition (XE) / Enterprise
Data Access Technology	ADODB (ActiveX Data Objects 2.8 Library)
OLE DB Provider Driver	Microsoft OLE DB Provider for Oracle (MSDAORA) / OraOLEDB
Covered Library Features	Single-Form Add Book, Issue Book, Return Book, DataGrid Display

1. Oracle 10g Database Setup (Table DDL Script)

Open **SQL*Plus** or **Oracle SQL Developer**, connect to your user schema (e.g., system or scott), and run the following script to create the library table:

```
-- Step 1.1: Create the library table
CREATE TABLE library (
    book_id      NUMBER(6) PRIMARY KEY,
    book_name    VARCHAR2(100) NOT NULL,
    author_name  VARCHAR2(100),
    quantity     NUMBER(4) DEFAULT 0
);

-- Step 1.2: Insert initial sample records for laboratory testing
INSERT INTO library VALUES (101, 'Data Structures', 'Mark Allen', 5);
INSERT INTO library VALUES (102, 'Operating Systems', 'Galvin', 3);
INSERT INTO library VALUES (103, 'Computer Networks', 'Tanenbaum', 7);

-- Step 1.3: Commit transaction to disk
COMMIT;
```

2. Visual Basic 6.0 Project Configuration & References

Before placing controls, add the required ADODB library and DataGrid component to your VB6 project:

Step 2.1 (Add ADO Reference): From the top menu, go to Project > References... Scroll down and check 'Microsoft ActiveX Data Objects 2.8 Library' (or version 2.5 / 2.6). Click OK. This provides the ADODB.Connection and ADODB.Recordset objects.

Step 2.2 (Add DataGrid Component): Go to Project > Components... (or press Ctrl + T). In the Controls list, check 'Microsoft DataGrid Control 6.0 (OLEDB)'. Click OK. The DataGrid icon will now appear in your Toolbox palette.

3. Form Design & Control Property Settings

In the Form Designer (**Form1.frm**), arrange the Frame, 4 Labels, 4 TextBoxes, 4 CommandButtons, and 1 DataGrid as shown below:

Control Type	Property Name	Value to Set	Functional Description
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Form	(Name) / Caption	Form1 / Library System	Main single-form interface window
Frame	Caption	Library Details	Groups input controls together
TextBox	(Name) / Text	txtBookId / (Empty)	Accepts numeric Book ID (PK)
TextBox	(Name) / Text	txtBookName / (Empty)	Accepts Book Title
TextBox	(Name) / Text	txtAuthor / (Empty)	Accepts Author Name
TextBox	(Name) / Text	txtQuantity / (Empty)	Stock count or issue/return count
CommandButton	(Name) / Caption	btnAddBook / Add Book	Executes SQL INSERT into Oracle
CommandButton	(Name) / Caption	btnIssueBook / Issue Book	Decrements quantity where qty >= issue
CommandButton	(Name) / Caption	btnReturnBook / Return Book	Increments quantity for Book ID
CommandButton	(Name) / Caption	btnRefresh / Refresh / View	Re-executes SELECT query to reload DataGrid

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CommandButton	(Name) / Caption	btnRefresh / Refresh / View	Re-executes SELECT query to reload DataGrid
DataGrid	(Name)	DataGrid1	OleDb DataGrid bound to client-side Recordset

4. Oracle 10g ADODB Data Connection

In classic VB6, an ADODB Connection is opened using the Microsoft OLE DB Provider for Oracle (**MSDAORA**) or the Oracle Provider (**OraOLEDB.Oracle**):

```
' Connection string format for Oracle 10g Express Edition (XE):
con.ConnectionString = "Provider=MSDAORA;Data Source=XE;User Id=system;Password=oracle;"

' Note: If using Oracle's native driver:
' con.ConnectionString = "Provider=OraOLEDB.Oracle;Data Source=XE;User
Id=system;Password=oracle;"
```

5. Complete Pure VB6 Source Code (Form1.frm)

Double-click **Form1** in VB6, press Ctrl + A, delete existing lines, and paste the complete pure VB6 code below:

```
Option Explicit

' Global ADODB Connection and Recordset Objects
Dim con As ADODB.Connection
Dim rs As ADODB.Recordset

' -----
' 1. Helper: Open Database Connection
' -----
Private Sub OpenConnection()
    If con Is Nothing Then
        Set con = New ADODB.Connection
    End If
    If con.State = adStateClosed Then
        ' Connect to Oracle 10g XE using MSDAORA OLE DB provider
        con.ConnectionString = "Provider=MSDAORA;Data Source=XE;User
Id=system;Password=oracle;"
        con.Open
    End If
End Sub

' -----
' 2. Helper: Fetch Records and Populate DataGrid1
' -----
Private Sub RefreshGrid()
    On Error GoTo ErrHandler
    Call OpenConnection

    If Not (rs Is Nothing) Then
        If rs.State = adStateOpen Then rs.Close
    End If

    Set rs = New ADODB.Recordset
    ' Client-side cursor is mandatory for binding to DataGrid control
    rs.CursorLocation = adUseClient
    rs.Open "SELECT book_id AS [Book ID], book_name AS [Book Name], " & _
        "author_name AS [Author Name], quantity AS [Quantity] " & _
        "FROM library ORDER BY book_id ASC", con, adOpenStatic, adLockOptimistic

    ' Bind recordset to DataGrid1
    Set DataGrid1.DataSource = rs
    Exit Sub

ErrHandler:
    MsgBox "Error loading grid: " & Err.Description, vbCritical, "Database Error"
End Sub

' -----
' 3. Reusable Helper: Execute SQL INSERT / UPDATE queries
' -----
Private Sub ExecuteSQL(ByVal sql As String, ByVal successMsg As String)
    On Error GoTo ErrHandler
```



```

    Call OpenConnection

    Dim recordsAffected As Long
    con.Execute sql, recordsAffected

    If recordsAffected > 0 Then
        MsgBox successMsg, vbInformation, "Success"
        Call RefreshGrid
        ' Clear TextBoxes
        txtBookId.Text = ""
        txtBookName.Text = ""
        txtAuthor.Text = ""
        txtQuantity.Text = ""
        txtBookId.SetFocus
    Else
        MsgBox "Operation failed! Check Book ID or stock availability.", vbExclamation,
"Alert"
    End If
    Exit Sub

ErrorHandler:
    MsgBox "SQL Error: " & Err.Description, vbCritical, "Execution Error"
End Sub

' -----
' Form Load Event: Auto populate grid on startup
' -----
Private Sub Form_Load()
    Call RefreshGrid
End Sub

' -----
' BUTTON 1: ADD BOOK (INSERT)
' -----
Private Sub btnAddBook_Click()
    If Trim(txtBookId.Text) = "" Or Trim(txtBookName.Text) = "" Then
        MsgBox "Please enter Book ID and Book Name.", vbExclamation, "Validation"
        Exit Sub
    End If

    Dim sql As String
    sql = "INSERT INTO library (book_id, book_name, author_name, quantity) " & _
        "VALUES (" & Val(txtBookId.Text) & ", '" & Replace(txtBookName.Text, "'", "'')
& "', '" & _
        Replace(txtAuthor.Text, "'", "'') & "', " & Val(txtQuantity.Text) & ")"

    Call ExecuteSQL(sql, "Book Added Successfully!")
End Sub

' -----
' BUTTON 2: ISSUE BOOK (DECREASE QUANTITY)
' -----
Private Sub btnIssueBook_Click()
    If Trim(txtBookId.Text) = "" Then
        MsgBox "Please enter or click a Book ID to issue.", vbExclamation, "Validation"
        Exit Sub
    End If

```

```

    Dim qty As Long
    qty = Val(txtQuantity.Text)
    If qty <= 0 Then qty = 1

    Dim sql As String
    sql = "UPDATE library SET quantity = quantity - " & qty & _
        " WHERE book_id = " & Val(txtBookId.Text) & " AND quantity >= " & qty

    Call ExecuteSQL(sql, "Book Issued Successfully! Stock Updated.")
End Sub

' -----
' BUTTON 3: RETURN BOOK (INCREASE QUANTITY)
' -----
Private Sub btnReturnBook_Click()
    If Trim(txtBookId.Text) = "" Then
        MsgBox "Please enter or click a Book ID to return.", vbExclamation, "Validation"
        Exit Sub
    End If

    Dim qty As Long
    qty = Val(txtQuantity.Text)
    If qty <= 0 Then qty = 1

    Dim sql As String
    sql = "UPDATE library SET quantity = quantity + " & qty & _
        " WHERE book_id = " & Val(txtBookId.Text)

    Call ExecuteSQL(sql, "Book Returned Successfully! Stock Updated.")
End Sub

' -----
' BUTTON 4: REFRESH / VIEW
' -----
Private Sub btnRefresh_Click()
    Call RefreshGrid
End Sub

' -----
' DataGridView RowColChange: Auto-fill textboxes from selected grid row
' -----
Private Sub DataGridView1_RowColChange(LastRow As Variant, ByVal LastCol As Integer)
    On Error Resume Next
    If Not (rs Is Nothing) Then
        If rs.State = adStateOpen And Not rs.EOF And Not rs.BOF Then
            txtBookId.Text = rs.Fields("Book ID").Value
            txtBookName.Text = rs.Fields("Book Name").Value
            txtAuthor.Text = rs.Fields("Author Name").Value
            txtQuantity.Text = rs.Fields("Quantity").Value
        End If
    End If
End Sub

' -----
' Form Unload: Release memory and connections cleanly
' -----

```



```
Private Sub Form_Unload(Cancel As Integer)
    On Error Resume Next
    If Not (rs Is Nothing) Then
        If rs.State = adStateOpen Then rs.Close
        Set rs = Nothing
    End If
    If Not (con Is Nothing) Then
        If con.State = adStateOpen Then con.Close
        Set con = Nothing
    End If
End Sub
```

6. Visual Basic 6.0 Runtime Output Windows

The following screenshots illustrate the runtime behavior across all four main operations:

Figure 6.1: Initial Form Load & DataGrid Records Display

Library Management System - VB6 & Oracle 10g

Library Details

Book ID:

Author Name:

Book Name:

Quantity:

Add Book

Issue Book

Return Book

Refresh / View

Oracle 10g Records (DataGrid1):

	Book ID	Book Name	Author Name	Quantity	
	101	Data Structures	Mark Allen	5	
	102	Operating Systems	Galvin	3	
	103	Computer Networks	Tanenbaum	7	

Description: When Form1 loads, Form_Load triggers RefreshGrid(), opening an ADODB client-side recordset and binding it directly to DataGrid1.

Figure 6.2: Inserting a New Book Record (Add Book)

Library Management System - [Adding Record]

Library Details

Book ID:

104

Author Name:

Henry F. Korth

Book Name:

Database System Concepts

Quantity:

6

Add Book

Issue Book

Return Book

Refresh / View

Oracle 10g Records (DataGrid1)

Book ID	Book Name	Quantity
101	Data Structure	5
102	Operating Syst	3
103	Computer Netv	7

Success

i

Book Added Successfully!

OK

Description: Inputting Book ID 104 ('Database System Concepts', Author 'Henry F. Korth', Quantity 6) and clicking 'Add Book' executes SQL INSERT with success MsgBox.

Figure 6.3: Refreshed DataGrid After Addition

Library Management System - [DataGrid Refreshed]

Library Details

Book ID:

Author Name:

Book Name:

Quantity:

Add Book

Issue Book

Return Book

Refresh / View

Oracle 10g Records (DataGrid1):

Book ID	Book Name	Author Name	Quantity
101	Data Structures	Mark Allen	5
102	Operating Systems	Galvin	3
103	Computer Networks	Tanenbaum	7
▶ 104	Database System Concepts	Henry F. Korth	6

Description: After inserting, textboxes are cleared and DataGrid1 automatically refreshes to display the newly inserted record with Book ID 104.

Figure 6.4: Issuing a Book (Stock Decrement)

Library Management System - [Issue Book]

Library Details

Book ID:101

Author Name:Mark Allen

Book Name:Data Structures

Quantity:1

Add Book

Issue Book

Return Book

Refresh / View

Oracle 10g Records (DataGrid1)

Book ID	Book Name	Quantity
101	Data Structures	4
102	Operating Systems	3
103	Computer Networks	7
104	Database System Concepts	6

Description: Clicking row 101 auto-fills fields. Clicking 'Issue Book' executes UPDATE decrementing quantity from 5 to 4 in Oracle 10g.

Figure 6.5: Returning a Book (Stock Increment)

Library Management System - [Return Book]

Library Details

Book ID:102

Author Name:Galvin

Book Name:Operating Systems

Quantity:1

Add Book

Issue Book

Return Book

Refresh / View

Oracle 10g Records (DataGrid1)

Book ID	Book Name	Quantity
101	Data Structures	4
102	Operating Systems	4
103	Computer Networks	7
104	Database System Concepts	6

Description: Clicking row 102 and pressing 'Return Book' increments quantity from 3 to 4, restoring inventory in Oracle.